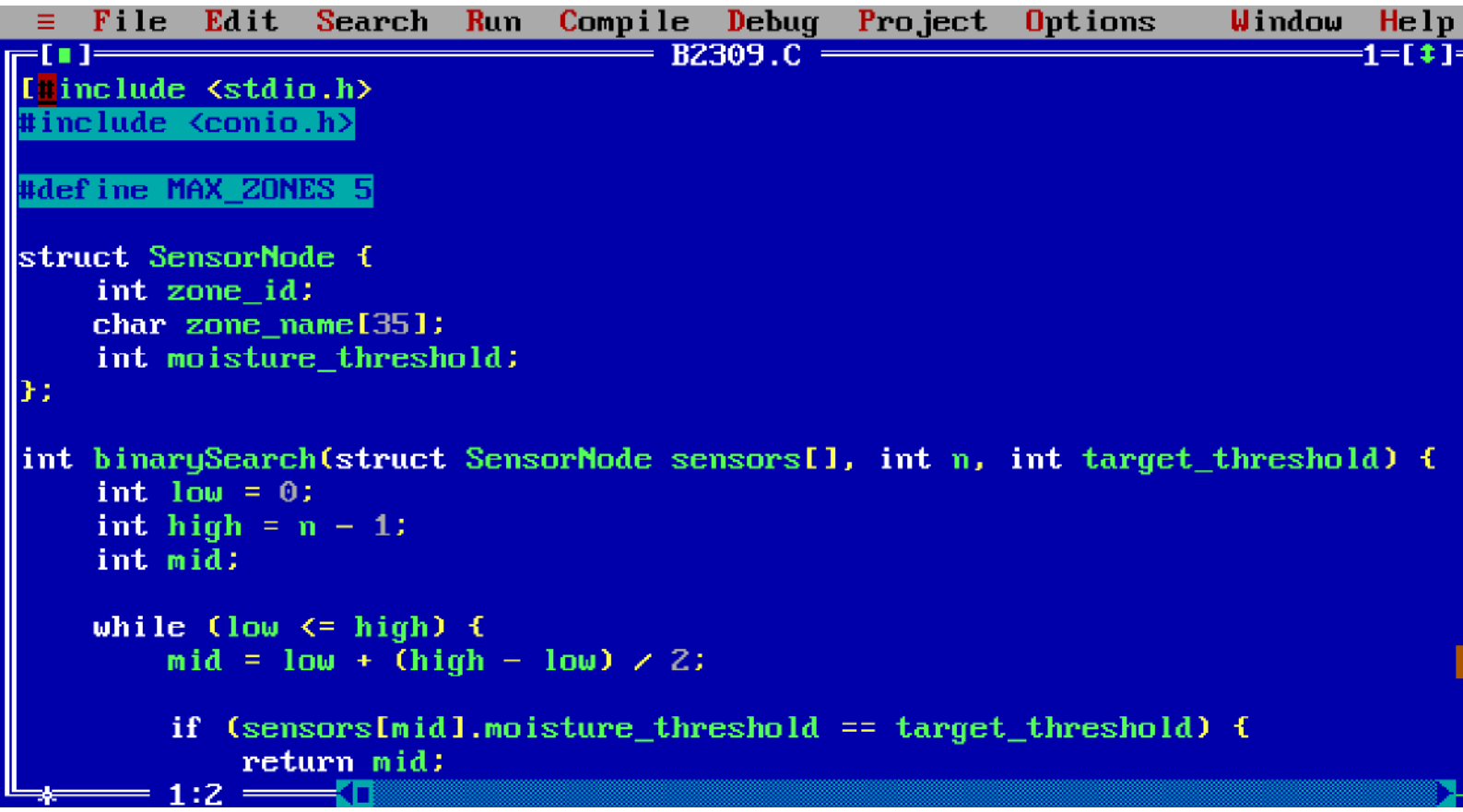

File Edit Search Run Compile Debug Project Options Window Help
B2309.C
#include <stdio.h>
#include <conio.h>
#define MAX_ZONES 5
struct SensorNode {
int zone_id;
char zone_name[35];
int moisture_threshold;
};
int binarySearch(struct SensorNode sensors[], int n, int target_threshold) {
int low = 0;
int high = n - 1;
int mid;
while (low <= high) {
mid = low + (high - low) / 2;
if (sensors[mid].moisture_threshold == target_threshold) {
return mid;
1:2

```
≡ File  Edit  Search  Run  Compile  Debug  Project  Options    Window  Help
[■]═══════════════════ B2309.C ═══════════════════1=[↕]
        }

        if (sensors[mid].moisture_threshold < target_threshold) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }

    return -1;
}

int main() {
    int i, search_val, result_index;
    struct SensorNode sensors[MAX_ZONES];

    clrscr();

    sensors[0].zone_id = 101;
    sensors[0].moisture_threshold = 15;
    sprintf(sensors[0].zone_name, "Zone B (Tillering Stage)");
 ═══ 42:27 ═══
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```

```
≡ File  Edit  Search  Run  Compile  Debug  Project  Options    Window  Hel
[■]═══════════════════ B2309.C ═══════════════════1=[↕
    sensors[1].zone_id = 102;
    sensors[1].moisture_threshold = 25;
    sprintf(sensors[1].zone_name, "Zone D (Ripening Stage)");

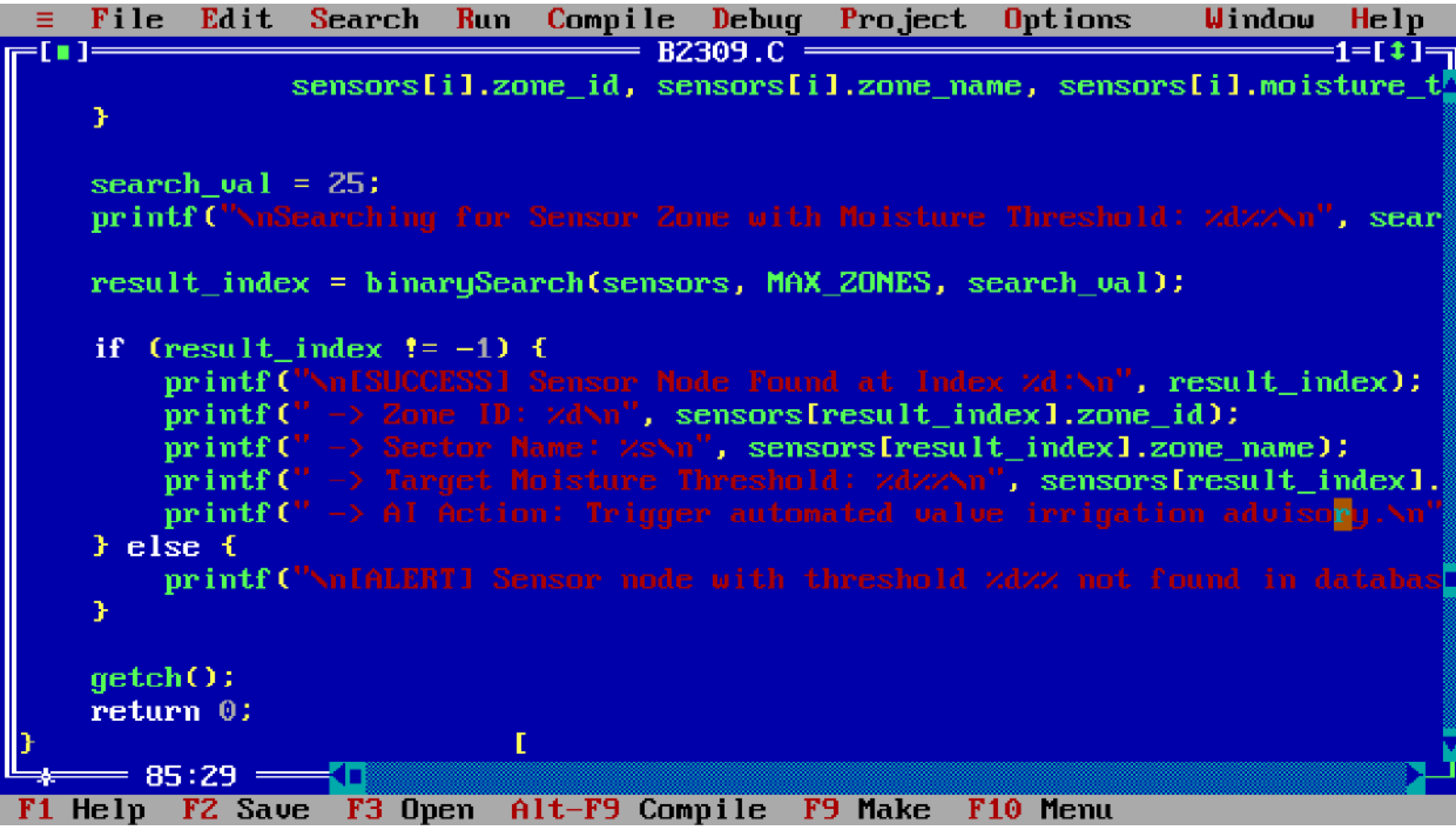
    sensors[2].zone_id = 103;
    sensors[2].moisture_threshold = 30;
    sprintf(sensors[2].zone_name, "Zone E (Sprouting Stage)");

    sensors[3].zone_id = 104;
    sensors[3].moisture_threshold = 40;
    sprintf(sensors[3].zone_name, "Zone A (Germination Stage)");

    sensors[4].zone_id = 105;
    sensors[4].moisture_threshold = 60;
    sprintf(sensors[4].zone_name, "Zone C (Grand Growth Stage)");

    printf("=== SUGARCANE IRRIGATION ADVISORY BINARY SEARCH ===\n\n");

    printf("Sorted Sensor Threshold Database:\n");
    for (i = 0; i < MAX_ZONES; i++) {
        printf(" -> [ID: %[d] %s - Threshold: %d%%\n",
═══ 64:28 ═══
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```


File Edit Search Run Compile Debug Project Options Window Help
B2309.C
sensors[i].zone_id, sensors[i].zone_name, sensors[i].moisture_t
}
search_val = 25;
printf("\nSearching for Sensor Zone with Moisture Threshold: %d%%\n", sear
result_index = binarySearch(sensors, MAX_ZONES, search_val);
if (result_index != -1) {
printf("\n[SUCCESS] Sensor Node Found at Index %d:\n", result_index);
printf(" -> Zone ID: %d\n", sensors[result_index].zone_id);
printf(" -> Sector Name: %s\n", sensors[result_index].zone_name);
printf(" -> Target Moisture Threshold: %d%%\n", sensors[result_index].
printf(" -> AI Action: Trigger automated valve irrigation advisory.\n"
} else {
printf("\n[ALERT] Sensor node with threshold %d%% not found in databas
}
getch();
return 0;
}
85:29
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

```
=== SUGARCANE IRRIGATION ADVISORY BINARY SEARCH ===

Sorted Sensor Threshold Database:
 -> [ID: 101] Zone B (Tillering Stage) - Threshold: 15%
 -> [ID: 102] Zone D (Ripening Stage) - Threshold: 25%
 -> [ID: 103] Zone E (Sprouting Stage) - Threshold: 30%
 -> [ID: 104] Zone A (Germination Stage) - Threshold: 40%
 -> [ID: 105] Zone C (Grand Growth Stage) - Threshold: 60%

Searching for Sensor Zone with Moisture Threshold: 25%

[SUCCESS] Sensor Node Found at Index 1:
 -> Zone ID: 102
 -> Sector Name: Zone D (Ripening Stage)
 -> Target Moisture Threshold: 25%
 -> AI Action: Trigger automated valve irrigation advisory.
```